

T0-Theory: Cosmic Relations

Contents

.1	Introduction: The Universal ξ -Constant	2
	Foundations of T0 Theory	2
	Time-Energy Duality as Foundation	2
.2	Cosmic Microwave Background (CMB)	3
	CMB without Big Bang: ξ -Field Mechanisms	3
	CMB Energy Density and ξ -Length Scale	3
.3	Casimir Effect and ξ -Field Connection	4
	Casimir-CMB Ratio	4
	ξ -Field as Universal Vacuum	4
.4	Cosmic Redshift without Expansion	5
	ξ -Field Mechanism	5
.5	Structure Formation in the Static ξ -Universe	5
	Continuous Structure Development	5
	ξ -Supported Continuous Creation	5
.6	Dimensionless ξ -Hierarchy	5
	Energy Scale Ratios	5
.7	Experimental Predictions and Tests	6
	Precision Measurements of Gravitational Constant	6
	Casimir Force Anomalies	6
	Electromagnetic Resonance	6
.8	Cosmological Consequences	7
	Cosmological Problems in the T0 Picture	7
	Parameter Reduction	7

Abstract

The T0-theory examines how far cosmic phenomena can be traced back to a single universal constant $\xi = \frac{4}{3} \times 10^{-4}$. This document presents the fundamental relationships between the gravitational constant, cosmic microwave background radiation (CMB), Casimir effect and cosmic structures within the framework of a static, eternally existing universe. All derivations are performed in natural units ($\hbar = c = k_B = 1$) and rest on the corpus's own time-energy reciprocity $T \cdot E = 1$ (Doc. 306).

Note on versions. The German version of this document is more extensive than this English one. Earlier revisions shortened or changed parts of the English version. Where the two differ, the more extensive German version applies.

Status markers. Statements marked **[S]** are *speculative*: a hint or conjecture that has not yet been derived or numerically checked in the corpus. This document deals almost exclusively with the cosmic sector (static universe, CMB, redshift, dark energy). The current corpus deliberately leaves this sector aside, because the Standard Model does not derive it either (P39); the cosmological conclusions of this document are therefore to be read as **[S]**.

.1 Introduction: The Universal ξ -Constant

Foundations of T0 Theory

Important

T0 theory is based on the universal dimensionless constant $\xi = \frac{4}{3} \times 10^{-4}$, to which it traces phenomena from the subatomic to the cosmic scale.

T0 theory is built on a single fundamental constant. This constant forms the basis for all physical calculations and predictions of the theory:

$$\xi = \frac{4}{3} \times 10^{-4} = 1.333333... \times 10^{-4} \quad (1)$$

This dimensionless constant connects quantum and gravitational phenomena and is intended as the basis of a unified description of the fundamental interactions.

Note on Derivation

For the detailed derivation and physical justification of this fundamental constant, see the document "Parameter Derivation" (available at:).

Time-Energy Duality as Foundation

Thesis

[S] A classical Big Bang singularity with infinite density is not physically realized; its place is taken by a tiny but finite core with minimal length scale L_0 (from ξ). The justification is provided by the corpus's own time-energy reciprocity $T \cdot E = 1$ (Doc. 306).

In natural units ($\hbar = c = k_B = 1$) the time-mass duality $T \cdot m = 1$ with $E = m$ yields the exact reciprocity:

$$T \cdot E = 1 \quad (2)$$

This has the following cosmological consequences **[S]**:

- From $T \cdot E = 1$ it follows that $E \rightarrow \infty \Leftrightarrow T \rightarrow 0$
- Time is bounded from below by $T_0 = \xi t_P$; hence E is bounded from above, $E \leq 1/T_0$
- An infinite energy density is structurally excluded: there is no singular beginning
- The torus is compact and boundaryless ($\partial T^4 = \emptyset$); the question of a beginning at $t = 0$ presupposes a boundary that it does not possess
- The universe is static, without expanding space

Heisenberg's uncertainty relation $\Delta E \times \Delta t \geq \frac{1}{2}$ does not carry this argument: for finite Δt it yields only a finite lower bound on ΔE , and its Δt is the characteristic time of change of an observable, not a cosmic age (Doc. 306).

.2 Cosmic Microwave Background (CMB)

CMB without Big Bang: ξ -Field Mechanisms

Thesis

[S] If time-energy duality excludes a Big Bang, as concluded above, the CMB must have a different origin than the $z=1100$ decoupling of standard cosmology.

T0-theory interprets the CMB **[S]** as radiation from ξ -field quantum fluctuations:

$$\frac{T_{\text{CMB}}}{E_{\xi}} = \frac{16}{9} \xi^2 \quad (3)$$

With $E_{\xi} = \frac{1}{\xi} = \frac{3}{4} \times 10^4$ (natural units) and $\xi = \frac{4}{3} \times 10^{-4}$ this yields:

$$T_{\text{CMB}} = \frac{16}{9} \xi^2 \times E_{\xi} = \frac{16}{9} \times 1.78 \times 10^{-8} \times 7500 = 2.37 \times 10^{-4} \quad (4)$$

Conversion to SI units:

$$T_{\text{CMB}} = 2.751 \text{ K} \quad (5)$$

With E_{ξ} in eV, the value 2.370×10^{-4} corresponds to 2.751 K; the measured value is 2.725 K. The expression thus lies about 1% from the observed value (cf. the table of dimensionless ratios: 3.13 versus 3.16×10^{-8}). This agreement is unit-dependent: the relation gives a pure number that agrees with the measured value only in the energy unit eV; it is not a derivation, and the question is open.

CMB Energy Density and ξ -Length Scale

The CMB energy density in natural units is:

$$\rho_{\text{CMB}} = \frac{\pi^2}{15} T_{\text{CMB}}^4 \approx 2.0 \times 10^{-15} \text{ eV}^4 \quad (\text{natural units, dimension } [E^4]) \quad (6)$$

This energy density defines a characteristic ξ -length scale:

$$L_{\xi} = \left(\frac{\xi}{\rho_{\text{CMB}}} \right)^{1/4} \quad (7)$$

Fundamental relation of CMB energy density:

$$\rho_{\text{CMB}} = \frac{\xi}{L_{\xi}^4} = \frac{\frac{4}{3} \times 10^{-4}}{(L_{\xi})^4} \quad (8)$$

.3 Casimir Effect and ξ -Field Connection

Casimir-CMB Ratio

Experiment

The ratio between Casimir energy density and CMB energy density at the characteristic ξ -length scale $L_\xi = 10^{-4}$ m is an identity, because L_ξ itself is determined from the measured CMB energy density and ξ ; it is calculated, not a measured value.

The Casimir energy density at plate separation $d = L_\xi$ is:

$$|\rho_{\text{Casimir}}| = \frac{\pi^2}{720 \times L_\xi^4} \quad (\text{natural units}) \quad (9)$$

The formula $\pi^2 \hbar c / (240 d^4)$ describes the Casimir pressure between ideal plates; the energy density is $\pi^2 \hbar c / (720 d^4)$.

The ratio yields:

$$\frac{|\rho_{\text{Casimir}}|}{\rho_{\text{CMB}}} = \frac{\pi^2}{720 \xi} = \frac{\pi^2 \times 10^4}{960} \approx 102.8 \quad (10)$$

Direct calculation with SI values: With $L_\xi = 10^{-4}$ m, direct calculation gives:

$$|\rho_{\text{Casimir}}| = \frac{\hbar c \pi^2}{720 \times (10^{-4})^4} = 4.33 \times 10^{-12} \text{ J/m}^3 \quad (11)$$

$$\rho_{\text{CMB}} = 4.17 \times 10^{-14} \text{ J/m}^3 \quad (12)$$

$$\text{Ratio} = \frac{4.33 \times 10^{-12}}{4.17 \times 10^{-14}} = 104 \quad (13)$$

The ratio is calculated, not a measured value; only ρ_{CMB} is measured. The theoretical expression (102.8) and the direct calculation with the measured ρ_{CMB} (104) differ by about 1%; this is the same formula with a rounded L_ξ .

ξ -Field as Universal Vacuum

Important

The ξ -field manifests both in free CMB radiation and in geometrically constrained Casimir vacuum. In the interpretation of this document this is a statement about scales **[S]**, not evidence for the physical reality of the ξ -field.

At the characteristic ξ -length scale L_ξ , the CMB vacuum energy density and the Casimir energy density do not coincide (ratio about 10^2); equality lies at a larger plate separation:

$$\text{Free vacuum: } \rho_{\text{CMB}} \approx 2.0 \times 10^{-15} \text{ eV}^4 \quad (14)$$

$$\text{Constrained vacuum: } |\rho_{\text{Casimir}}| = \frac{\pi^2}{720 d^4} \quad (15)$$

.4 Cosmic Redshift without Expansion

ξ -Field Mechanism

Thesis

[S] According to this thesis, the observed cosmic redshift arises not from spatial expansion but from fractal path lengthening in the omnipresent ξ -field.

The redshift is achromatic (K6): the fractal path lengthening is purely geometric and stretches all wavelengths by the same factor, $1 + z = e^{\xi x}$, independent of λ ; the scale of the path lengthening is calibrated via H_0 (P39).

An energy-dependent energy loss of the photons, such as $dE/dx = -\xi E^2/E_\xi$, does not hold, because it would give a wavelength-dependent redshift; the observed cosmological redshift is achromatic.

.5 Structure Formation in the Static ξ -Universe

Continuous Structure Development

In the static T0 universe **[S]**, structure formation occurs continuously without Big Bang constraints:

$$\frac{d\rho}{dt} = -\nabla \cdot (\rho \mathbf{v}) + S_\xi(\rho, T, \xi) \quad (16)$$

where S_ξ is the ξ -field source term for continuous matter/energy transformation.

ξ -Supported Continuous Creation

The ξ -field enables continuous matter/energy transformation:

$$\text{Quantum vacuum} \xrightarrow{\xi} \text{Virtual particles} \quad (17)$$

$$\text{Virtual particles} \xrightarrow{\xi^2} \text{Real particles} \quad (18)$$

$$\text{Real particles} \xrightarrow{\xi^3} \text{Atomic nuclei} \quad (19)$$

$$\text{Atomic nuclei} \xrightarrow{\text{Time}} \text{Stars, galaxies} \quad (20)$$

Energy balance is maintained by:

$$\rho_{\text{total}} = \rho_{\text{matter}} + \rho_{\xi\text{-field}} = \text{constant} \quad (21)$$

.6 Dimensionless ξ -Hierarchy

Energy Scale Ratios

The ξ -relations used here reduce to exact mathematical ratios:

Table 1: Dimensionless ξ -ratios

Ratio	Expression	Value
Temperature	$\frac{T_{\text{CMB}}}{E_\xi}$	3.13×10^{-8}
Relation (unit-dependent)	$\frac{16}{9}\xi^2$	3.16×10^{-8}
Length	$\frac{\ell_\xi}{L_\xi}$	$\xi^{-1/4}$
Casimir-CMB	$\frac{ \rho_{\text{Casimir}} }{\rho_{\text{CMB}}}$	$\frac{\pi^2 \times 10^4}{960}$

Important

The ξ -relations used here consist of exact mathematical ratios:

- Fractions: $\frac{4}{3}$, $\frac{3}{4}$, $\frac{16}{9}$
- Powers of ten: 10^{-4} , 10^3 , 10^4
- Mathematical constants: π^2

No arbitrary decimal numbers occur in these expressions; they are built from ξ and integer structural choices (motivated settings).

.7 Experimental Predictions and Tests

Precision Measurements of Gravitational Constant

T0-theory predicts:

$$G_{\text{T0}} = 6.6745 \times 10^{-11} \text{ m}^3/(\text{kg} \cdot \text{s}^2) \quad (+0.003 \% \text{ relative to CODATA}) \quad (22)$$

The corpus formula $G = \xi^2/(4m_e) \cdot C_{\text{conv}} \cdot K_{\text{frak}}$ yields 6.6745×10^{-11} . C_{conv} is the SI conversion of the single-anchor chain (Doc. 383); its numerical value 7.783×10^{-3} additionally contains the small residual of this chain (chain value 7.58×10^{-3}). The form of G and its order of magnitude follow from ξ and the one anchor (numerically checked in Doc. 383); with the anchor m_e , G lies at -2.6% , with v at -0.46% (Doc. 383). The agreement to 0.003% arises only with this residual and does not count as an independent precision result.

Casimir Force Anomalies

Experiment

Prediction: Casimir force anomalies at characteristic ξ -length scale

- Standard Casimir law: $F \propto d^{-4}$
- ξ -field modifications at $d = L_\xi = 10^{-4} \text{ m}$
- Measurable deviations through ξ -vacuum coupling

Electromagnetic Resonance

Maximum ξ -field-photon coupling at characteristic frequency:

$$\nu_\xi = \frac{c}{L_\xi} \approx 3 \times 10^{12} \text{ Hz} = 3 \text{ THz} \quad (23)$$

Here $1/L_\xi = 10^4 \text{ m}^{-1}$ is a wavenumber; the frequency is c/L_ξ .
Electromagnetic anomalies should occur at this frequency.

.8 Cosmological Consequences

Cosmological Problems in the T0 Picture

The following table shows how known problems of standard cosmology appear in the static T0 model **[S]**. The entries are interpretations in the cosmic sector, which the current corpus leaves aside (P39), not derived solutions:

Table 2: Cosmological problems: Standard vs. T0

Problem	Λ CDM	T0 Interpretation
Horizon problem	Inflation required	Infinite causal connectivity
Flatness problem	Fine-tuning	Geometry stabilizes over infinite time
Monopole problem	Topological defects	Defects dissipate over infinite time
Lithium problem	Nucleosynthesis discrepancy	Nucleosynthesis over unlimited time
Age problem	Objects older than universe	Objects can be arbitrarily old
H_0 tension	9% discrepancy	No H_0 in static universe
Dark energy	69% of energy density	Not required

Parameter Reduction

Parameter count

Parameter count: the Standard Model and Λ CDM together use 25+ parameters, T0-theory the single parameter ξ (plus integer structural choices as motivated settings; SI anchors serve only for conversion).

- Standard model of particle physics: 19+ parameters
- Λ CDM cosmology: 6 parameters
- T0-theory: 1 parameter (ξ)

Arithmetically this corresponds to a 96% reduction. The Λ CDM share concerns the cosmic sector, which the current corpus leaves aside (P39).

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